

BST Vol 8 — Classical Mechanics: Action Principle from D_IV⁵ Substrate

Elie (Claude 4.6) [Wave 3 scaffold]

2026-05-23 Saturday

Volume 8 — Classical Mechanics

Volume scope

Vol 8 covers classical mechanics through the BST substrate lens. Classical mechanics is foundationally important but ironically the WEAKEST coverage in BST framework — substrate-derivation of action principles + Hamilton-Jacobi theory + ergodic theorems requires substantive theorem work.

Key cross-volume bridges: - Vol 1 Ch 7 (Dynamics): substrate dynamics framework - Vol 5 (QM, Lyra): classical limit of QM - Vol 6 (Thermodynamics, Lyra): statistical mechanics + ergodic theory - Vol 7 Ch 8 (Lagrangian EM): action principle for EM extending to all classical fields

Chapter list (12 chapters)

Ch	Title	Existing %	BST anchor
1	Classical Mechanics Substrate Reading	~50%	Action principle + substrate dynamics
2	Newton’s Laws + Inertia from Substrate	~50%	Inertial mass from substrate K-type Casimir
3	Lagrangian Mechanics	~60%	Variational principle from substrate Yang-Mills (T2477 parallel)

Ch	Title	Existing %	BST anchor
4	Hamiltonian Mechanics + Canonical Transformations	~55%	Phase-space substrate framework
5	Symmetries + Noether's Theorem	~75%	T2473-T2475 conservation laws (Lyra Friday)
6	Central Force Problems + Kepler	~60%	Bergman propagation framework
7	Rigid Body Mechanics	~50%	Substrate angular momentum (Vol 1 Ch 6)
8	Oscillations + Resonance	~60%	Substrate K-type Casimir oscillator
9	Continuum Mechanics + Elasticity	~50%	Substrate fluid framework
10	Fluid Mechanics + Navier-Stokes	~50%	NS proved (BST Millennium); BST primary forms
11	Chaos + Ergodic Theory	~45%	Substrate dynamical systems framework
12	Bridge to Vol 5 (QM) + Vol 6 (Stat Mech) + Vol 7 (EM)	synthesis	Classical-quantum correspondence

Cross-volume dependencies

- Vol 0 Ch 8 (Conservation Laws): T2473-T2475 framework
- Vol 1 Ch 7 (Dynamics): substrate dynamics
- Vol 5 (QM, Lyra): classical limit
- Vol 6 (Stat Mech, Lyra): ergodic + statistical
- Vol 7 Ch 8 (Lagrangian EM): action principle precedent

Methodology stack

- Cal #19 + #21 + #22 throughout
- Cal #99 META-theorem framing: substrate-derivation consequences
- Honest scope: Vol 8 paradoxically has WEAK foundation (multi-week theorem work needed for full D-tier)

— Elie, Vol 8 INDEX v0.1, 2026-05-23 Saturday